SNx4HC04 Hex Inverters

1 Features

- Buffered inputs
- Wide operating voltage range: 2 V to 6 V
- Wide operating temperature range: –40°C to +85°C
- Supports fanout up to 10 LSTTL loads
- Significant power reduction compared to LSTTL logic ICs

2 Applications

- Synchronize invterted clock inputs
- Debounce a switch
- Invert a digital signal

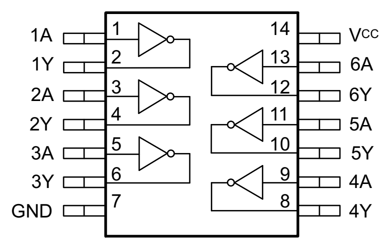
3 Description

This device contains six independent inverters. Each gate performs the Boolean function $Y = \overline{A}$ in positive logic.

Device Information(1)

PART NUMBER	PACKAGE	BODY SIZE (NOM)
SN74HC04D	SOIC (14)	8.70 mm × 3.90 mm
SN74HC04DB	SSOP (14)	6.50 mm × 5.30 mm
SN74HC04N	PDIP (14)	19.30 mm × 6.40 mm
SN74HC04NS	SO (14)	10.20 mm × 5.30 mm
SN74HC04PW	TSSOP (14)	5.00 mm × 4.40 mm
SN54HC04J	CDIP (14)	21.30 mm × 7.60 mm
SN54HC04W	CFP (14)	9.20 mm × 6.29 mm
SN54HC04FK	LCCC (20)	8.90 mm × 8.90 mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.



Functional pinout

5 Pin Configuration and Functions

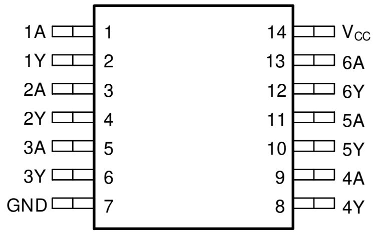


Figure 5-1. D, DB, N, NS, PW, J, or W Package 14-Pin SOIC, SSOP, PDIP, SO, TSSOP, CDIP, or CFP ph Top View

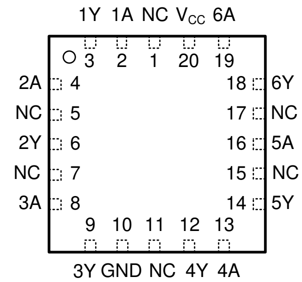


Figure 5-2. FK Package 20-Pin LCCC Top View

Pin Functions

PIN			I/O	DESCRIPTION
NAME	D, DB, N, NS, PW, J, or W	FK		
1A	1	2	Input	Channel 1, Input A
1Y	2	3	Output	Channel 1, Output Y
2A	3	4	Input	Channel 2, Input A
2Y	4	6	Output	Channel 2, Output Y
3A	5	8	Input	Channel 3, Input A
3Y	6	9	Output	Channel 3, Output Y
GND	7	10	—	Ground
4Y	8	12	Output	Channel 4, Output Y
4A	9	13	Input	Channel 4, Input A
5Y	10	14	Output	Channel 5, Output Y
5A	11	16	Input	Channel 5, Input A
6Y	12	18	Output	Channel 6, Output Y
6A	13	19	Input	Channel 6, Input A
V_{CC}	14	20	—	Positive Supply
NC		1, 5, 7, 11, 15, 17	—	Not internally connected

6 Specifications

6.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)[(1)]

			MIN	MAX	UNIT
V_{CC}	Supply voltage		–0.5	7	V
I_{IK}	Input clamp current[(2)]	$V_I < 0$ or $V_I > V_{CC}$		±20	mA
I_{OK}	Output clamp current[(2)]	$V_O < 0$		±20	mA
I_O	Continuous output current	$V_O = 0$ to V_{CC}		±25	mA
	Continuous current through V_{CC} or GND			±50	mA
T_J	Junction temperature[(3)]			150	°C
T_{stg}	Storage temperature		–60	150	°C

(1) Stresses beyond those listed under *Absolute Maximum Rating* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Condition*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
(2) The input and output voltage ratings may be exceeded if the input and output current ratings are observed.
(3) Guaranteed by design.

6.2 ESD Ratings

			VALUE	UNIT
$V_{(ESD)}$	Electrostatic discharge	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001[(1)]	±2000	V
		Charged-device model (CDM), per JEDEC specification JESD22-C101[(2)]	–	

(1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.
(2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

6.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)

			MIN	NOM	MAX	UNIT
V_{CC}	Supply voltage		2	5	6	V
V_{IH}	High-level input voltage	V_{CC} = 2 V	1.5			V
		V_{CC} = 4.5 V	3.15			
		V_{CC} = 6 V	4.2			
V_{IL}	Low-level input voltage	V_{CC} = 2 V			0.5	V
		V_{CC} = 4.5 V			1.35	
		V_{CC} = 6 V			1.8	
V_I	Input voltage		0		V_{CC}	V
V_O	Output voltage		0		V_{CC}	V
$\Delta t/\Delta v$	Input transition rise and fall rate	V_{CC} = 2 V			1000	ns
		V_{CC} = 4.5 V			500	
		V_{CC} = 6 V			400	
T_A	Operating free-air temperature	SN54HC04	–55		125	°C
		SN74HC04	–40		85	

6.4 Thermal Information

THERMAL METRIC[(1)]		SN74HC04					UNIT
		D (SOIC)	DB (SSOP)	N (PDIP)	NS (SOP)	PW (TSSOP)	
		14 PINS	14 PINS	14 PINS	14 PINS	14 PINS	
$R_{\theta JA}$	Junction-to-ambient thermal resistance	133.6	114.8	60.7	122.6	151.7	°C/W
$R_{\theta JC(top)}$	Junction-to-case (top) thermal resistance	89	64.5	47.8	81.8	79.4	°C/W
$R_{\theta JB}$	Junction-to-board thermal resistance	89.5	65.1	40.6	83.8	94.7	°C/W
Ψ_{JT}	Junction-to-top characterization parameter	45.5	23.7	26.9	45.4	25.2	°C/W
Ψ_{JB}	Junction-to-board characterization parameter	89.1	64.4	40.3	83.4	94.1	°C/W
$R_{\theta JC(bot)}$	Junction-to-case (bottom) thermal resistance	N/A	N/A	N/A	N/A	N/A	°C/W

(1) For more information about traditional and new thermal metrics, see the Semiconductor and IC Package Thermal Metrics application report.

6.5 Electrical Characteristics - 74

over operating free-air temperature range; typical values measured at T_A = 25°C (unless otherwise noted).

PARAMETER		TEST CONDITIONS		V_{CC}	Operating free-air temperature (T_A)						UNIT
					25°C			-40°C to 85°C			
					MIN	TYP	MAX	MIN	TYP	MAX	
V_{OH}	High-level output voltage	$V_I = V_{IH}$ or V_{IL}	I_{OH} = –20 µA	2 V	1.9	1.998		1.9			V
				4.5 V	4.4	4.499		4.4			
				6 V	5.9	5.999		5.9			
			I_{OH} = –4 mA	4.5 V	3.98	4.3		3.84			
			I_{OH} = –5.2 mA	6 V	5.48	5.8		5.34			
V_{OL}	Low-level output voltage	$V_I = V_{IH}$ or V_{IL}	I_{OL} = 20 µA	2 V		0.002	0.1			0.1	V
				4.5 V		0.001	0.1			0.1	
				6 V		0.001	0.1			0.1	
			I_{OL} = 4 mA	4.5 V		0.17	0.26			0.33	
			I_{OL} = 5.2 mA	6 V		0.15	0.26			0.33	
I_I	Input leakage current	$V_I = V_{CC}$ or 0		6 V			±0.1			±1	µA
I_{CC}	Supply current	$V_I = V_{CC}$ or 0	$V_I = V_{CC}$ or 0	6 V			2			20	µA
C_i	Input capacitance			2 V to 6 V		3	10			10	pF

6.6 Electrical Characteristics - 54

over operating free-air temperature range; typical values measured at TA = 25°C (unless otherwise noted).

PARAMETER		TEST CONDITIONS		V_{CC}	Operating free-air temperature (T_A)									UNIT
					25°C			–40°C to 85°C			–55°C to 125°C			
					MIN	TYP	MAX	MIN	TYP	MAX	MIN	TYP	MAX	
V_{OH}	High-level output voltage	$V_I = V_{IH}$ or V_{IL}	I_{OH} = –20 µA	2 V	1.9	1.998		1.9			1.9			V
				4.5 V	4.4	4.499		4.4			4.4			
				6 V	5.9	5.999		5.9			5.9			
			I_{OH} = –4 mA	4.5 V	3.98	4.3		3.84			3.7			
			I_{OH} = –5.2 mA	6 V	5.48	5.8		5.34			5.2			
V_{OL}	Low-level output voltage	$V_I = V_{IH}$ or V_{IL}	I_{OL} = 20 µA	2 V		0.002	0.1			0.1			0.1	V
				4.5 V		0.001	0.1			0.1			0.1	
				6 V		0.001	0.1			0.1			0.1	
			I_{OL} = 4 mA	4.5 V		0.17	0.26			0.33			0.4	
			I_{OL} = 5.2 mA	6 V		0.15	0.26			0.33			0.4	
I_I	Input leakage current	$V_I = V_{CC}$ or 0		6 V			±0.1			±1			±1	µA
I_{CC}	Supply current	$V_I = V_{CC}$ or 0	I_O = 0	6 V			2			20			40	µA
C_i	Input capacitance			2 V to 6 V		3	10			10			10	pF

6.7 Switching Characteristics - 74

over operating free-air temperature range (unless otherwise noted)

PARAMETER		FROM	TO	V_{CC}	Operating free-air temperature (T_A)						UNIT
					25°C			–40°C to 85°C			
					MIN	TYP	MAX	MIN	TYP	MAX	
t_{pd}	Propagation delay	A	Y	2 V		45	95			120	ns
				4.5 V		9	19			24	
				6 V		8	16			20	
t_t	Transition-time		Y	2 V		38	75			95	ns
				4.5 V		8	15			19	
				6 V		6	13			16	

6.8 Switching Characteristics - 54

over operating free-air temperature range; typical values measured at TA = 25°C (unless otherwise noted).

PARAMETER		FROM	TO	V_{CC}	Operating free-air temperature (T_A)									UNIT
					25°C			–40°C to 85°C			–55°C to 125°C			
					MIN	TYP	MAX	MIN	TYP	MAX	MIN	TYP	MAX	
t_{pd}	Propagation delay	A	Y	2 V		45	95			120			125	ns
				4.5 V		9	19			24			29	
				6 V		8	16			20			25	
t_t	Transition-time		Y	2 V		38	75			95			110	ns
				4.5 V		8	15			19			22	
				6 V		6	13			16			19	